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Summary

Third-year Computer Science undergraduate specializing in AI & Machine Learning at Arka Jain University (CGPA: 8.20/10) with production internship experience engineering leak-free ML and tool-orchestration pipelines at Venturing Digitally. Builds end-to-end intelligent systems ranging from severe class-imbalance classifiers (resolving 578:1 fraud skews via in-pipeline SMOTE and PR sweeps) to multi-agent architectures orchestrating 5 specialized security agents over WebSockets. Enforces defensive runtime verification across generative models, deploying programmatic numeric-grounding guards over BigQuery invoice records and eliminating a 66% fake-citation reward-hacking exploit in LangGraph loops. Leads the Microsoft Learn Student Ambassador campus chapter and founded Students for Startups, delivering technical Azure AI workshops for 150+ student engineers.

Experience

- AI/ML Engineer Intern** | *Venturing Digitally Pvt. Ltd.* May 2026 – Jul 2026
- Eliminated data leakage and resolved extreme class imbalance across tabular and text datasets by isolating preprocessing inside cross-validation folds and tuning tree ensemble thresholds via PR sweeps.
 - Prevented runtime orchestration failures by engineering deterministic tool routing, Pydantic schema validation, and fail-safe defensive error boundaries for production LLM workflows.
- Microsoft Learn Student Ambassador (MLSA)** | *Microsoft (Campus Lead)* 2026 – Present
- Connected student engineers with early-stage tech founders by founding “Students for Startups”, facilitating collaborative mentorship and technical feedback.
 - Trained 150+ engineering students on Microsoft Azure AI, cloud infrastructure, and agentic pipelines by leading hands-on technical workshops.

Projects

- DecisionForge** | [GitHub](#) | *React 19, Google BigQuery, Node.js, Express 5, Google Gemini, Python* 2026
- Eliminated LLM financial hallucination risks in GST tax audits by engineering a numeric-grounding verifier that validates generated notices against raw BigQuery invoice tables.
 - Audited Section 16(2)(aa) 18% annual interest exposure across 4 risk tiers by developing BigQuery SQL views that reconcile purchase registers into 7 invoice categories with INR 100 tolerance.
- AgentGuard** | [GitHub](#) | *FastAPI, Python, Azure OpenAI (GPT-4o), Azure AI Search, WebSockets, Docker, React* 2026
- Automated incident triage and webhook containment across 3 attack simulation scenarios (SQL injection, credential stuffing, insider exfiltration) by orchestrating 5 specialized agents.
 - Achieved zero paid cloud dependencies by building a lightweight typed async Python agent framework with local in-memory fallbacks and real-time WebSocket telemetry to a React dashboard.
- Credit Card Fraud Detection Pipeline** | [GitHub](#) | *Python, scikit-learn, XGBoost, imbalanced-learn, SMOTE, Streamlit* 2026
- Lifted fraud detection precision from 36.1% to 87.5% at 78.6% recall on 284,807 transactions by calibrating the XGBoost operating threshold to 0.9793 along the PR-AUC curve (0.8506).
 - Neutralized severe 578:1 class imbalance (0.17% fraud rate) without test-set contamination by encapsulating cyclical time features, RobustScaler, and SMOTE inside an imblearn pipeline.
- ResearchMind** | [GitHub](#) | *Python, LangGraph, Mistral, Tavily API, BeautifulSoup, Pydantic, Streamlit* 2026
- Eliminated a 66% fake-citation reward-hacking exploit in multi-agent feedback loops by enforcing programmatic Pydantic score caps (< 7) until claims match retrieved sources.
 - Guaranteed fact-grounded report synthesis by architecting a 4-agent LangGraph workflow that automatically routes detected knowledge gaps back to live web search rather than ungrounded LLM rewrites.

Technical Skills

Machine Learning & Data Science: Python, scikit-learn, XGBoost, LightGBM, imbalanced-learn (SMOTE), pandas, NumPy, joblib, Feature Engineering, Threshold Calibration
Generative AI & Agentic Systems: LangGraph, LangChain, Google Gemini, Azure OpenAI, RAG, ChromaDB, Pydantic, Tool Routing, Multi-Agent Orchestration
Backend & Cloud Systems: FastAPI, PostgreSQL, SQLite, Docker, Docker Compose, WebSockets, RESTful APIs, Git, Linux
Developer Tools & Frontend: pytest, Streamlit, TypeScript, React, Vite, Tailwind CSS

Education

Arka Jain University Jamshedpur, India
Bachelor of Technology (B.Tech) in Computer Science (AI & Machine Learning) | CGPA: 8.20/10 2024 – 2028

Achievements & Certifications

- **Google Cloud APAC Gen AI Hackathon:** Ranked 78th out of hundreds of teams across Asia.
- **Certifications:** DeepLearning.AI (LangChain for LLM Application Development), Google Cloud (Generative AI Leader), Microsoft Certified (Azure AI Fundamentals), IBM (AI Engineering Professional).